

Damien Kyle Harwood

Full-Stack Developer · TypeScript, React, Node

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Full-stack developer based in Durban with three years of hands-on experience building and maintaining production software using TypeScript, React and Node.js.

For the past two years I have worked as the sole developer within an ISP and its group companies, taking ownership of approximately ten production systems across a wide range of business requirements. That has given me strong exposure to the full software development lifecycle: gathering requirements from non-technical teams, designing and building the solution, deploying it, and then troubleshooting, maintaining and supporting it in production.

My work has covered internal APIs and shared services, customer-facing web applications, third-party system integrations, reporting and business intelligence, and application security. I have integrated eleven third-party APIs in production, most of which behave differently to their documentation, and I have designed access control from first principles across a frontend and an API.

Working independently has required me to become highly adaptable, self-motivated and comfortable taking ownership of complex problems. I have developed my skills primarily through real-world projects, tight deadlines and continuous self-learning.

TypeScript is my strongest technical skill, with practical experience across both frontend and backend development.

WHAT I WORK WITH

Strongest Solid	TypeScript, JavaScript, React, Next.js, Node.js, Express, REST API design MySQL, MongoDB (Mongoose), SQL, schema design, migrations, JWT auth, role-based access control, Git
Also work with	Docker, containerised deploys, CI/CD, Vite, Tailwind, Drizzle ORM, nginx, AWS
Security	OWASP Top 10 remediation, RBAC design, JWT claims, bcrypt, CSP and HSTS, CVE triage
AI / LLM	Model Context Protocol servers, self-hosted models, LLM-assisted classification
Earlier	C#, ASP.NET MVC, Blazor, Entity Framework, Unity

EXPERIENCE

Full-Stack Developer

Apr 2025 – present

UrbanX Connect / FirstNet (First Technology Group), Durban

Sole developer across roughly ten production systems for an ISP and its group companies, covering architecture, delivery, deployment, security and third-party vendor relationships.

- Designed and built a shared internal API in TypeScript, Express and Mongoose to replace logic that had been duplicated across every project. A core internal application went from **over 30 seconds to under 5**, and one route from 10s to 0.92s. A fix now lands once rather than being re-implemented per repository.
- Designed and built role-based access control from first principles across a Next.js frontend and an Express API. Database-backed roles with per-section grants, enforced in middleware and again in the per-section server layouts, plus welcome-email and forgot-password flows on single-use tokens with no account enumeration.
- Built an end-to-end automation pipeline that replaced a recurring manual process consuming most of a working day: email ingestion, classification, lifecycle detection, duplicate suppression, matching against live records, LLM-assisted geographic resolution constrained to a known place-name list, and reviewable drafts published to a public status page. Released behind a switch so that nothing could reach a customer unintentionally.
- Remediated OWASP Top 10 defect classes in production systems, including broken access control, privilege escalation, insecure direct object reference, mass assignment, ReDoS in user-supplied filters and account enumeration on password reset. Also diagnosed a double-hashing defect in password reset that had been silently locking users out after every administrative reset.
- Re-architected a database estate off a single shared credential onto per-service isolated databases behind SSH tunnels, so that a compromise of one service cannot reach the others. Delivered with no downtime.
- Built business intelligence dashboards over live operational data, including geographic density and target-area ranking, and a signup funnel instrumented to separate the distinct reasons a signup stalls. The findings changed how the business targeted its advertising spend.

- Rewrote how the estate logs. Reduced 197 loose console calls to 45 behind a single helper, and closed 67 empty catch blocks that had been quietly swallowing real failures behind bare 500s.
- Responsible for the deployment estate across a dozen services: Docker, containers, reverse proxy, DNS, certificates and secrets, with no downtime on container swaps. Also migrated network monitoring to a new platform on vendor decommission, preserving a live customer-facing feature.
- Integrated eleven third-party APIs in production, including billing and CRM platforms, credit bureaux, payment gateways, telephony and messaging APIs, status pages and advertising APIs. Carried one billing platform through a major version upgrade and a sandbox restoration.
- Ran requirements sessions directly with finance, sales and operations leads to map manual workflows before automating them, and write up the root cause of every incident, including the ones I caused myself.

Remote Engineer

Jan 2024 – Mar 2025

First Technology, remote

- Ran software deployments to devices nationally, and helped bring on the junior members of a team of six.
- Project-managed two client projects end to end, which largely meant keeping other people unblocked and the timelines realistic.

Junior Game Developer

Aug 2023 – Dec 2023

SUN, remote

- Built and maintained several of the company's games in Construct 3 and JavaScript.

Site Reliability Engineer (part-time)

Dec 2022 – Jun 2023

Bitcast, remote

- Maintained client-facing systems and servers using Python, Django, AWS, Kubernetes, Docker, Terraform and Grafana. My first proper exposure to infrastructure as something to be designed rather than tolerated.

EDUCATION

BCIS, Computer and Information Sciences (Game Design and Development)

2022

IIE Vega School

The degree is in game development. Everything since, including web development, database design, infrastructure and application security, I have taught myself and applied directly in production.

ADDITIONAL

Location	Durban. Open to Cape Town, Johannesburg, hybrid and remote.
Notice	One calendar month, in writing.
Also at	Codewars · LeetCode
References	On request.